MIRA: Longitudinal Analysis

Dynamic and reproducible statistical report

2026-10-25

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4 Descriptive summary **42**

1 Abstract

This report presents the observed analytical results for symptom.

2 Analysis context

2.1 Resolved configuration

Id: case_id

Id generated: FALSE

2.1.1 Outcome names

Symptom

Key: symptom

Declared: symptom

Source: outcomes

Label: SYMPTOM

2.1.2 Time vars

Symptom: pre, week4, final

2.1.3 Time labels

Symptom: Baseline, Week 4, Week 12

Improvement direction: lower

Stable threshold: 1

Variable pattern: auto

2.1.4 Auto detected

Id: FALSE

Time vars: FALSE

Time labels: FALSE

Arm: TRUE

Reference arm: TRUE

Covariates: FALSE

Improvement direction: FALSE

Stable threshold: FALSE

2.1.5 Specified manually

Id: TRUE

Time vars: TRUE

Time labels: TRUE

Arm: FALSE

Reference arm: FALSE

Covariates: TRUE

2.1.6 Sources

Id: manual

Arm: none

Improvement direction: user

Stable threshold: user

2.1.7 Analyses

Core: descriptives, missingness, change, variability, trajectories

Figures: TRUE

Primary mixed-effects model: TRUE

Diagnostic outliers: TRUE

Correlations across time points: TRUE

Arm tests: TRUE

2.2 Dataset overview

N rows: 42

N columns: 4

2.2.1 Column profile

Table 1: Column profile

variable	class	numeric	unique n	missing n	missing pct
case_id	character	FALSE	42	0	0
pre	numeric	TRUE	42	0	0
week4	numeric	TRUE	42	0	0
final	numeric	TRUE	42	0	0

3 Outcome: SYMPTOM

3.1 Analysis settings

Alpha: 0.05

Confidence level: 0.95

Confidence percent: 95

P adjust method: holm

Improvement direction: lower

Stable threshold: 1

Arm tests: FALSE

Strict id: TRUE

Non finite handling: Inf/-Inf treated as NA for analysis

Time vars: pre, week4, final

Time labels: Baseline, Week 4, Week 12

3.2 Overview

Outcome: symptom

Outcome display: SYMPTOM

Id: case_id

N rows: 42

N patients: 42

N timepoints: 3

Timepoints: pre, week4, final

Time labels: Baseline, Week 4, Week 12

Supplied time vars: TRUE

Missing ids: 0

Duplicated ids: 0

Complete profiles: 42

Complete profiles pct: 100

Non finite values: 0

Missing arm: 0

3.3 Missing data

3.3.1 By time point

Table 2: By time point (block 1 of 2)

time	label	missing n	missing pct	non finite n	non finite pct	unavailable n
pre	Baseline	0	0	0	0	0
week4	Week 4	0	0	0	0	0
final	Week 12	0	0	0	0	0

Table 3: By time point (block 2 of 2)

time	label	unavailable pct
pre	Baseline	0
week4	Week 4	0
final	Week 12	0

3.3.2 By participant

Table 4: By participant

patient	missing n	non finite n	unavailable n	unavailable pct
E001	0	0	0	0
E002	0	0	0	0
E003	0	0	0	0
E004	0	0	0	0
E005	0	0	0	0
E006	0	0	0	0
E007	0	0	0	0
E008	0	0	0	0
E009	0	0	0	0
E010	0	0	0	0
E011	0	0	0	0
E012	0	0	0	0
E013	0	0	0	0
E014	0	0	0	0
E015	0	0	0	0
E016	0	0	0	0
E017	0	0	0	0
E018	0	0	0	0
E019	0	0	0	0
E020	0	0	0	0
E021	0	0	0	0
E022	0	0	0	0
E023	0	0	0	0
E024	0	0	0	0
E025	0	0	0	0
E026	0	0	0	0
E027	0	0	0	0
E028	0	0	0	0
E029	0	0	0	0
E030	0	0	0	0
E031	0	0	0	0
E032	0	0	0	0
E033	0	0	0	0
E034	0	0	0	0

patient	missing n	non finite n	unavailable n	unavailable pct
E035	0	0	0	0
E036	0	0	0	0
E037	0	0	0	0
E038	0	0	0	0
E039	0	0	0	0
E040	0	0	0	0
E041	0	0	0	0
E042	0	0	0	0

3.4 Descriptive statistics

Table 5: Descriptive statistics (block 1 of 5)

time	label	n	missing	missing pct	non finite	non finite pct	unavailable
pre	Baseline	42	0	0	0	0	0
week4	Week 4	42	0	0	0	0	0
final	Week 12	42	0	0	0	0	0

Table 6: Descriptive statistics (block 2 of 5)

time	label	unavailable pct	symptom mean	symptom sd	symptom variance
pre	Baseline	0	29.982	5.385	28.999
week4	Week 4	0	27.789	5.365	28.780
final	Week 12	0	26.175	5.777	33.378

Table 7: Descriptive statistics (block 3 of 5)

time	label	symptom se	symptom ci lower	symptom ci upper	symptom median
pre	Baseline	0.831	28.304	31.660	29.520
week4	Week 4	0.828	26.117	29.461	27.266
final	Week 12	0.891	24.375	27.976	26.244

Table 8: Descriptive statistics (block 4 of 5)

time	label	symptom q1	symptom q3	symptom iqr	symptom min
pre	Baseline	27.264	33.764	6.500	19.466
week4	Week 4	25.149	31.123	5.974	16.585
final	Week 12	23.121	30.197	7.076	12.668

Table 9: Descriptive statistics (block 5 of 5)

time	label	symptom max	symptom cv percent
pre	Baseline	41.653	17.961
week4	Week 4	38.269	19.305
final	Week 12	39.465	22.072

3.5 Within-participant changes

Table 10: Within-participant changes (block 1 of 6)

from	from label	to	to label	n	symptom mean from	symptom mean to	symptom mean change
pre	Baseline	week4	Week 4	42	29.982	27.789	-2.193
pre	Baseline	final	Week 12	42	29.982	26.175	-3.806
week4	Week 4	final	Week 12	42	27.789	26.175	-1.613

Table 11: Within-participant changes (block 2 of 6)

from	from label	symptom sd change	symptom se change	symptom ci lower	symptom ci upper
pre	Baseline	1.082	0.167	-2.530	-1.856
pre	Baseline	1.539	0.238	-4.286	-3.327
week4	Week 4	2.004	0.309	-2.238	-0.989

Table 12: Within-participant changes (block 3 of 6)

from	from label	symptom cohens dz	symptom increased n	symptom increased pct	symptom decreased n
pre	Baseline	-2.027	0	0.000	36
pre	Baseline	-2.473	0	0.000	42
week4	Week 4	-0.805	6	14.286	27

Table 13: Within-participant changes (block 4 of 6)

from	from label	symptom decreased pct	symptom stable n	symptom stable pct	symptom improved n
pre	Baseline	85.714	6	14.286	36
pre	Baseline	100.000	0	0.000	42
week4	Week 4	64.286	9	21.429	27

Table 14: Within-participant changes (block 5 of 6)

from	from label	symptom improved pct	symptom worsened n	symptom worsened pct	symptom paired t p
pre	Baseline	85.714	0	0.000	<0.001
pre	Baseline	100.000	0	0.000	<0.001
week4	Week 4	64.286	6	14.286	<0.001

Table 15: Within-participant changes (block 6 of 6)

from	from label	symptom paired t p adj	symptom wilcoxon p	symptom wilcoxon p adj
pre	Baseline	<0.001	<0.001	<0.001
pre	Baseline	<0.001	<0.001	<0.001
week4	Week 4	<0.001	<0.001	<0.001

3.6 Individual trajectories

Table 16: Individual trajectories (block 1 of 2)

patient	outcome	arm	symptom baseline	symptom final	symptom absolute change	symptom relative change percent
E001	symptom	–	23.909	21.228	-2.681	-11.214
E002	symptom	–	27.151	24.542	-2.609	-9.610
E003	symptom	–	32.987	31.084	-1.903	-5.768
E004	symptom	–	29.309	25.755	-3.553	-12.124
E005	symptom	–	32.888	30.675	-2.213	-6.728
E006	symptom	–	27.750	24.980	-2.770	-9.982
E007	symptom	–	34.878	30.560	-4.318	-12.380
E008	symptom	–	27.602	22.856	-4.745	-17.192
E009	symptom	–	30.745	26.268	-4.476	-14.560
E010	symptom	–	19.466	16.273	-3.193	-16.404
E011	symptom	–	33.908	29.916	-3.992	-11.773
E012	symptom	–	36.668	32.406	-4.261	-11.621
E013	symptom	–	30.500	26.777	-3.723	-12.205
E014	symptom	–	31.131	29.132	-1.998	-6.420
E015	symptom	–	27.145	24.636	-2.509	-9.242
E016	symptom	–	37.868	34.674	-3.194	-8.434
E017	symptom	–	32.919	27.411	-5.508	-16.731
E018	symptom	–	28.925	23.327	-5.597	-19.352
E019	symptom	–	32.869	28.149	-4.720	-14.361
E020	symptom	–	19.935	13.441	-6.494	-32.575
E021	symptom	–	33.331	30.069	-3.263	-9.789
E022	symptom	–	41.653	39.465	-2.188	-5.254
E023	symptom	–	28.053	23.278	-4.774	-17.019
E024	symptom	–	24.676	19.390	-5.286	-21.421
E025	symptom	–	26.095	24.481	-1.614	-6.185
E026	symptom	–	34.383	30.240	-4.143	-12.049
E027	symptom	–	36.083	33.147	-2.936	-8.136
E028	symptom	–	28.411	26.813	-1.598	-5.625
E029	symptom	–	29.485	22.642	-6.843	-23.207
E030	symptom	–	29.555	26.219	-3.336	-11.287
E031	symptom	–	34.464	31.903	-2.561	-7.431
E032	symptom	–	27.616	23.897	-3.719	-13.466
E033	symptom	–	30.774	27.441	-3.333	-10.830
E034	symptom	–	24.984	22.526	-2.458	-9.837
E035	symptom	–	28.818	23.069	-5.749	-19.949
E036	symptom	–	39.109	36.235	-2.874	-7.349
E037	symptom	–	22.551	21.146	-1.405	-6.229
E038	symptom	–	20.055	14.893	-5.162	-25.738
E039	symptom	–	36.118	28.969	-7.149	-19.793
E040	symptom	–	28.864	26.005	-2.859	-9.904
E041	symptom	–	36.087	30.774	-5.313	-14.722
E042	symptom	–	19.513	12.668	-6.845	-35.080

Table 17: Individual trajectories (block 2 of 2)

patient	outcome	direction	clinical direction
E001	symptom	Decrease	Improved
E002	symptom	Decrease	Improved
E003	symptom	Decrease	Improved
E004	symptom	Decrease	Improved
E005	symptom	Decrease	Improved
E006	symptom	Decrease	Improved
E007	symptom	Decrease	Improved
E008	symptom	Decrease	Improved

patient	outcome	direction	clinical direction
E009	symptom	Decrease	Improved
E010	symptom	Decrease	Improved
E011	symptom	Decrease	Improved
E012	symptom	Decrease	Improved
E013	symptom	Decrease	Improved
E014	symptom	Decrease	Improved
E015	symptom	Decrease	Improved
E016	symptom	Decrease	Improved
E017	symptom	Decrease	Improved
E018	symptom	Decrease	Improved
E019	symptom	Decrease	Improved
E020	symptom	Decrease	Improved
E021	symptom	Decrease	Improved
E022	symptom	Decrease	Improved
E023	symptom	Decrease	Improved
E024	symptom	Decrease	Improved
E025	symptom	Decrease	Improved
E026	symptom	Decrease	Improved
E027	symptom	Decrease	Improved
E028	symptom	Decrease	Improved
E029	symptom	Decrease	Improved
E030	symptom	Decrease	Improved
E031	symptom	Decrease	Improved
E032	symptom	Decrease	Improved
E033	symptom	Decrease	Improved
E034	symptom	Decrease	Improved
E035	symptom	Decrease	Improved
E036	symptom	Decrease	Improved
E037	symptom	Decrease	Improved
E038	symptom	Decrease	Improved
E039	symptom	Decrease	Improved
E040	symptom	Decrease	Improved
E041	symptom	Decrease	Improved
E042	symptom	Decrease	Improved

3.7 Correlations across time points

3.7.1 Pearson correlation

Table 18: Pearson correlation

row	pre	week4	final
pre	1.000	0.980	0.964
week4	0.980	1.000	0.938
final	0.964	0.938	1.000

3.7.2 Spearman correlation

Table 19: Spearman correlation

row	pre	week4	final
pre	1.000	0.965	0.945
week4	0.965	1.000	0.912
final	0.945	0.912	1.000

3.7.3 Pairwise sample sizes

Table 20: Pairwise sample sizes

row	pre	week4	final
pre	42	42	42
week4	42	42	42
final	42	42	42

3.8 Variability and ICC

Table 21: Variability and ICC (block 1 of 3)

outcome	symptom grand mean	symptom between subject sd	symptom within subject sd	symptom ICC
symptom	27.982	5.436	1.811	0.959

Table 22: Variability and ICC (block 2 of 3)

outcome	symptom grand mean	symptom ICC model	symptom ICC anova complete profiles	symptom ms between
symptom	27.982	0.959	0.851	88.639

Table 23: Variability and ICC (block 3 of 3)

outcome	symptom grand mean	symptom ms within	symptom df between	symptom df within
symptom	27.982	4.88	41	84

3.9 Primary mixed-effects model

3.9.1 Fitted model object

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: model_formula
   Data: model_data

REML criterion at convergence: 563.1

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.11768 -0.46997 -0.01717  0.50696  1.62860

Random effects:
 Groups         Name        Variance Std.Dev.
 patient_factor (Intercept) 29.13    5.397
 Residual                    1.26    1.122
Number of obs: 126, groups:  patient_factor, 42

Fixed effects:
                  Estimate Std. Error      df t value Pr(>|t|)
(Intercept)         29.9816     0.8506 43.3461  35.249  < 2e-16 ***
time_factorWeek 4   -2.1929     0.2449 82.0000  -8.954 8.89e-14 ***
time_factorWeek 12  -3.8063     0.2449 82.0000 -15.542  < 2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) tm_fW4
tim_fctrWk4 -0.144
tm_fctrWk12 -0.144  0.500
```

3.9.2 Model ANOVA table

Table 24: Model ANOVA table

Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
306.595	153.298	2	82	121.705	<0.001

3.9.3 Global time test

Table 25: Global time test (block 1 of 2)

npar	AIC	BIC	logLik	-2*log(L)	Chisq
3	684.057	692.565	-339.028	678.057	–
5	572.274	586.455	-281.137	562.274	115.783

Table 26: Global time test (block 2 of 2)

npar	AIC	Df	Pr(>Chisq)
3	684.057	–	–
5	572.274	2	<0.001

Singular: FALSE

Converged: TRUE

Fixed-effect parameters: 3

3.10 Advanced longitudinal tests

3.10.1 Estimated marginal means and contrasts

Time

Object

```
 time_factor emmean    SE   df lower.CL upper.CL
 Baseline      30.0 0.851 43.4     28.3     31.7
 Week 4        27.8 0.851 43.4     26.1     29.5
 Week 12       26.2 0.851 43.4     24.5     27.9

Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95
```

Tidy

Table 27: Tidy (block 1 of 2)

time factor	emmean	SE	df	lower CL
Baseline	29.982	0.851	43.346	28.267
Week 4	27.789	0.851	43.346	26.074
Week 12	26.175	0.851	43.346	24.460

Table 28: Tidy (block 2 of 2)

time factor	emmean	upper CL
Baseline	29.982	31.697
Week 4	27.789	29.504
Week 12	26.175	27.890

Pairwise

```
 contrast           estimate    SE df t.ratio p.value
 Baseline - Week 4      2.19 0.245 82   8.954 <0.0001
 Baseline - Week 12     3.81 0.245 82  15.542 <0.0001
 Week 4 - Week 12       1.61 0.245 82   6.588 <0.0001

Degrees-of-freedom method: kenward-roger
```

Pairwise tidy

Table 29: Pairwise tidy (block 1 of 4)

contrast	symptom estimate	symptom SE	symptom df	symptom lower CL
Baseline - Week 4	2.193	0.245	82	1.706
Baseline - Week 12	3.806	0.245	82	3.319
Week 4 - Week 12	1.613	0.245	82	1.126

Table 30: Pairwise tidy (block 2 of 4)

contrast	symptom upper CL	symptom t ratio	symptom p raw	symptom p primary
Baseline - Week 4	2.680	8.954	<0.001	<0.001
Baseline - Week 12	4.293	15.542	<0.001	<0.001
Week 4 - Week 12	2.101	6.588	<0.001	<0.001

Table 31: Pairwise tidy (block 3 of 4)

contrast	symptom p bonferroni	symptom p holm	symptom p bh	symptom p by
Baseline - Week 4	<0.001	<0.001	<0.001	<0.001
Baseline - Week 12	<0.001	<0.001	<0.001	<0.001
Week 4 - Week 12	<0.001	<0.001	<0.001	<0.001

Table 32: Pairwise tidy (block 4 of 4)

contrast	primary method
Baseline - Week 4	holm
Baseline - Week 12	holm
Week 4 - Week 12	holm

Pairwise tukey

Table 33: Pairwise tukey (block 1 of 2)

contrast	estimate	SE	df	lower CL
Baseline - Week 4	2.193	0.245	82	1.608
Baseline - Week 12	3.806	0.245	82	3.222
Week 4 - Week 12	1.613	0.245	82	1.029

Table 34: Pairwise tukey (block 2 of 2)

contrast	upper CL	t ratio	p value
Baseline - Week 4	2.777	8.954	<0.001
Baseline - Week 12	4.391	15.542	<0.001
Week 4 - Week 12	2.198	6.588	<0.001

Baseline followup

Object

```
 contrast           estimate    SE df t.ratio p.value
 Week 4 - Baseline     -2.19 0.245 82  -8.954 <0.0001
 Week 12 - Baseline    -3.81 0.245 82 -15.542 <0.0001

Degrees-of-freedom method: kenward-roger
```

Tidy

Table 35: Tidy (block 1 of 4)

contrast	symptom estimate	symptom SE	symptom df	symptom lower CL
Week 4 - Baseline	-2.193	0.245	82	-2.680
Week 12 - Baseline	-3.806	0.245	82	-4.293

Table 36: Tidy (block 2 of 4)

contrast	symptom upper CL	symptom t ratio	symptom p raw	symptom p primary
Week 4 - Baseline	-1.706	-8.954	<0.001	<0.001
Week 12 - Baseline	-3.319	-15.542	<0.001	<0.001

Table 37: Tidy (block 3 of 4)

contrast	symptom p bonferroni	symptom p holm	symptom p bh	symptom p by
Week 4 - Baseline	<0.001	<0.001	<0.001	<0.001
Week 12 - Baseline	<0.001	<0.001	<0.001	<0.001

Table 38: Tidy (block 4 of 4)

contrast	primary method
Week 4 - Baseline	holm
Week 12 - Baseline	holm

Dunnett

Table 39: Dunnett (block 1 of 2)

contrast	estimate	SE	df	lower CL
Week 4 - Baseline	-2.193	0.245	82	-2.747
Week 12 - Baseline	-3.806	0.245	82	-4.360

Table 40: Dunnett (block 2 of 2)

contrast	upper CL	t ratio	p value
Week 4 - Baseline	-1.639	-8.954	<0.001
Week 12 - Baseline	-3.252	-15.542	<0.001

Dunnett method: emmeans dunnettx approximation

Baseline final

Object

```
 contrast         estimate    SE df t.ratio p.value
 final - baseline    -3.81 0.245 82 -15.542 <0.0001

Degrees-of-freedom method: kenward-roger
```

Tidy

Table 41: Tidy (block 1 of 4)

contrast	symptom estimate	symptom SE	symptom df	symptom lower CL
final - baseline	-3.806	0.245	82	-4.293

Table 42: Tidy (block 2 of 4)

contrast	symptom estimate	symptom upper CL	symptom t ratio	symptom p raw
final - baseline	-3.806	-3.319	-15.542	<0.001

Table 43: Tidy (block 3 of 4)

contrast	symptom estimate	symptom p primary	symptom p bonferroni	symptom p holm
final - baseline	-3.806	<0.001	<0.001	<0.001

Table 44: Tidy (block 4 of 4)

contrast	symptom estimate	symptom p bh	symptom p by	primary method
final - baseline	-3.806	<0.001	<0.001	holm

Consecutive

Object

```
 contrast          estimate    SE df t.ratio p.value
 Week 4 - Baseline    -2.19 0.245 82  -8.954 <0.0001
 Week 12 - Week 4     -1.61 0.245 82  -6.588 <0.0001

Degrees-of-freedom method: kenward-roger
```

Tidy

Table 45: Tidy (block 1 of 4)

contrast	symptom estimate	symptom SE	symptom df	symptom lower CL
Week 4 - Baseline	-2.193	0.245	82	-2.680
Week 12 - Week 4	-1.613	0.245	82	-2.101

Table 46: Tidy (block 2 of 4)

contrast	symptom upper CL	symptom t ratio	symptom p raw	symptom p primary
Week 4 - Baseline	-1.706	-8.954	<0.001	<0.001
Week 12 - Week 4	-1.126	-6.588	<0.001	<0.001

Table 47: Tidy (block 3 of 4)

contrast	symptom p bonferroni	symptom p holm	symptom p bh	symptom p by
Week 4 - Baseline	<0.001	<0.001	<0.001	<0.001
Week 12 - Week 4	<0.001	<0.001	<0.001	<0.001

Table 48: Tidy (block 4 of 4)

contrast	primary method
Week 4 - Baseline	holm
Week 12 - Week 4	holm

Trends

Object

```
 contrast  estimate    SE df t.ratio p.value
 linear      -3.806 0.245 82 -15.542 <0.0001
 quadratic    0.579 0.424 82   1.366  0.1756

Degrees-of-freedom method: kenward-roger
```

Tidy

Table 49: Tidy (block 1 of 4)

contrast	symptom estimate	symptom SE	symptom df	symptom lower CL
linear	-3.806	0.245	82	-4.293
quadratic	0.579	0.424	82	-0.264

Table 50: Tidy (block 2 of 4)

contrast	symptom estimate	symptom upper CL	symptom t ratio	symptom p raw
linear	-3.806	-3.319	-15.542	<0.001
quadratic	0.579	1.423	1.366	0.176

Table 51: Tidy (block 3 of 4)

contrast	symptom estimate	symptom p primary	symptom p bonferroni	symptom p holm
linear	-3.806	<0.001	<0.001	<0.001
quadratic	0.579	0.176	0.351	0.176

Table 52: Tidy (block 4 of 4)

contrast	symptom estimate	symptom p bh	symptom p by	primary method
linear	-3.806	<0.001	<0.001	holm
quadratic	0.579	0.176	0.263	holm

Scale: ordered timepoint levels

Note: Polynomial contrasts use the order of timepoints; they do not assume equally spaced chronological time.

3.10.2 Repeated-measures ANOVA

Object

```
Anova Table (Type 3 tests)

Response: value
       Effect    df  MSE          F  pes p.value
1 time_factor 2, 82 1.26 121.70 *** .748   <.001
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1
```

Model summary

Type: III

Repeated: TRUE

Univariate tests

Table 53: Univariate tests

row	Sum Sq	num Df	Error SS	den Df	F value	Pr(>F)
(Intercept)	98656.359	1	3634.185	41	1113.017	<0.001
time_factor	306.595	2	103.286	82	121.705	<0.001

Pval adjustments

Table 54: Pval adjustments

row	GG eps	Pr(>F[GG])	HF eps	Pr(>F[HF])
time_factor	0.7	<0.001	0.717	<0.001

Sphericity tests

Table 55: Sphericity tests

row	Test statistic	p-value
time_factor	0.571	0

Tidy

Table 56: Tidy (block 1 of 4)

effect	symptom partial eta squared	symptom generalized eta squared	symptom statistic	symptom df num
time_factor	0.748	0.076	121.705	2

Table 57: Tidy (block 2 of 4)

effect	symptom partial eta squared	symptom df den	symptom p value	symptom df num gg
time_factor	0.748	82	<0.001	1.399

Table 58: Tidy (block 3 of 4)

effect	symptom partial eta squared	symptom df den gg	symptom p gg	symptom df num hf
time_factor	0.748	57.363	<0.001	1.433

Table 59: Tidy (block 4 of 4)

effect	symptom partial eta squared	symptom df den hf	symptom p hf
time_factor	0.748	58.768	<0.001

Tables

Uncorrected partial eta

Table 60: Uncorrected partial eta

num Df	den Df	MSE	F	pes	Pr(>F)
2	82	1.26	121.705	0.748	<0.001

Uncorrected generalized eta

Table 61: Uncorrected generalized eta

num Df	den Df	MSE	F	ges	Pr(>F)
2	82	1.26	121.705	0.076	<0.001

Greenhouse geisser

Table 62: Greenhouse geisser

num Df	den Df	MSE	F	pes	Pr(>F)
1.399	57.363	1.801	121.705	0.748	<0.001

Huynh feldt

Table 63: Huynh feldt

num Df	den Df	MSE	F	pes	Pr(>F)
1.433	58.768	1.758	121.705	0.748	<0.001

Sphericity

Mauchly

Table 64: Mauchly

row	Test statistic	p-value
time_factor	0.571	0

Corrections

Table 65: Corrections

row	GG eps	Pr(>F[GG])	HF eps	Pr(>F[HF])
time_factor	0.7	<0.001	0.717	<0.001

N subjects: 42

3.10.3 Friedman test

Test

```
    Friedman rank sum test

data:  as.matrix(complete)
Friedman chi-squared = 69.857, df = 2, p-value = 6.772e-16
```

Tidy

Table 66: Tidy

n	n timepoints	symptom kendalls w	symptom statistic	symptom df	symptom p raw
42	3	0.832	69.857	2	<0.001

Posthoc

Tests

Pre to week4

```
    Wilcoxon signed rank test with continuity correction

data:  y and x
V = 0, p-value = 1.709e-08
alternative hypothesis: true location shift is not equal to 0
```

Pre to final

```
    Wilcoxon signed rank test with continuity correction

data:  y and x
V = 0, p-value = 1.709e-08
alternative hypothesis: true location shift is not equal to 0
```

Week4 to final

```
    Wilcoxon signed rank test with continuity correction

data:  y and x
V = 119, p-value = 3.307e-05
alternative hypothesis: true location shift is not equal to 0
```

Tidy

Table 67: Tidy (block 1 of 3)

from	from label	to	to label	n	symptom rank biserial	symptom statistic	symptom p raw
pre	Baseline	week4	Week 4	42	-1.000	0	<0.001
pre	Baseline	final	Week 12	42	-1.000	0	<0.001
week4	Week 4	final	Week 12	42	-0.736	119	<0.001

Table 68: Tidy (block 2 of 3)

from	from label	symptom p primary	symptom p bonferroni	symptom p holm	symptom p bh
pre	Baseline	<0.001	<0.001	<0.001	<0.001
pre	Baseline	<0.001	<0.001	<0.001	<0.001
week4	Week 4	<0.001	<0.001	<0.001	<0.001

Table 69: Tidy (block 3 of 3)

from	from label	symptom p by	primary method
pre	Baseline	<0.001	holm
pre	Baseline	<0.001	holm

from	from label	symptom p by	primary method
week4	Week 4	<0.001	holm

3.11 Advanced longitudinal models

3.11.1 Random-slope mixed-effects model

Model formula

```
value ~ time_factor + (1 + time_ordinal | patient_factor)
<environment: 0x00000141ad764078>
```

Time scale: ordered timepoint index

Note: The random slope uses the ordered timepoint index and does not imply equally spaced chronological time.

Primary mixed-effects model

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: formula
   Data: data

REML criterion at convergence: 561.4

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.08172 -0.51588 -0.01393  0.50141  1.71779

Random effects:
 Groups         Name         Variance Std.Dev. Corr
 patient_factor (Intercept)  27.44696 5.2390
                time_ordinal  0.02517 0.1586   1.00
 Residual                     1.23440 1.1110
Number of obs: 126, groups:  patient_factor, 42

Fixed effects:
                   Estimate Std. Error      df t value Pr(>|t|)
(Intercept)         29.9816     0.8264 41.6001  36.281  < 2e-16 ***
time_factorWeek 4   -2.1929     0.2437 82.8315  -8.999 6.67e-14 ***
time_factorWeek 12  -3.8063     0.2473 76.0867 -15.389  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) tm_fW4
tim_fctrWk4 -0.048
tm_fctrWk12  0.050  0.508
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

Fixed effects

Table 70: Fixed effects (block 1 of 2)

term	symptom Estimate	symptom Std Error	symptom df	symptom t value
(Intercept)	29.982	0.826	41.600	36.281
time_factorWeek 4	-2.193	0.244	82.831	-8.999

term	symptom Estimate	symptom Std Error	symptom df	symptom t value
time_factorWeek 12	-3.806	0.247	76.087	-15.389

Table 71: Fixed effects (block 2 of 2)

term	symptom Pr(>|t|)
(Intercept)	<0.001
time_factorWeek 4	<0.001
time_factorWeek 12	<0.001

Fixed effect tests

Table 72: Fixed effect tests

Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
294.674	147.337	2	78.93	119.36	<0.001

Variance components

Table 73: Variance components

grp	var1	var2	vcov	sdcor
patient_factor	(Intercept)	–	27.447	5.239
patient_factor	time_ordinal	–	0.025	0.159
patient_factor	(Intercept)	time_ordinal	0.831	1.000
Residual	–	–	1.234	1.111

Symptom random intercept slope correlation: 1

Symptom AIC: 575.429705650601

Symptom BIC: 595.283678999261

Symptom logLik: -280.7148528253

Converged: FALSE

Singular: TRUE

Global time test

Table 74: Global time test (block 1 of 2)

npar	AIC	BIC	logLik	-2*log(L)	Chisq
5	655.367	669.548	-322.683	645.367	–
7	574.600	594.454	-280.300	560.600	84.766

Table 75: Global time test (block 2 of 2)

npar	AIC	Df	Pr(>Chisq)
5	655.367	–	–
7	574.600	2	<0.001

Errors

Random effects comparison: all models must be fit to the same data object

3.11.2 NLME models

Compound-symmetry correlation

Model formula

```
value ~ time_factor
<environment: 0x00000141ad785a50>
```

Correlation structure: compound symmetry

Primary mixed-effects model

```
Linear mixed-effects model fit by REML
  Data: data
  Log-restricted-likelihood: -281.5317
  Fixed: fixed_formula
       (Intercept)  time_factorWeek 4 time_factorWeek 12
         29.981632          -2.192893          -3.806289

Random effects:
 Formula: ~1 | patient_factor
        (Intercept) Residual
StdDev:    5.396884 1.122313

Correlation Structure: Compound symmetry
 Formula: ~1 | patient_factor
 Parameter estimate(s):
Rho
  0
Number of Observations: 126
Number of Groups: 42
```

Model summary

```
Linear mixed-effects model fit by REML
  Data: data
       AIC      BIC    logLik
  575.0634 591.9366 -281.5317

Random effects:
 Formula: ~1 | patient_factor
        (Intercept) Residual
StdDev:    5.396884 1.122313

Correlation Structure: Compound symmetry
 Formula: ~1 | patient_factor
 Parameter estimate(s):
Rho
  0
Fixed effects:  list(fixed_formula)
                       Value Std.Error DF   t-value p-value
(Intercept)        29.981632 0.8505732 82  35.24874       0
time_factorWeek 4  -2.192893 0.2449088 82  -8.95391       0
time_factorWeek 12 -3.806289 0.2449088 82 -15.54166       0
 Correlation:
                   (Intr) tm_fW4
time_factorWeek 4  -0.144
time_factorWeek 12 -0.144  0.500
```

```
Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-2.11768002 -0.46997408 -0.01716623  0.50696332  1.62860013

Number of Observations: 126
Number of Groups: 42
```

Coefficients

Table 76: Coefficients (block 1 of 2)

row	Value	Std Error	DF	t-value
(Intercept)	29.982	0.851	82	35.249
time_factorWeek 4	-2.193	0.245	82	-8.954
time_factorWeek 12	-3.806	0.245	82	-15.542

Table 77: Coefficients (block 2 of 2)

row	p-value
(Intercept)	0
time_factorWeek 4	0
time_factorWeek 12	0

Symptom correlation parameter: 0

Symptom AIC: 575.063448570921

Symptom BIC: 591.936554703156

Symptom logLik: -281.531724285461

Converged: TRUE

AR(1) correlation

Model formula

```
value ~ time_factor
<environment: 0x00000141ad785a50>
```

Correlation structure: AR(1)

Time scale: ordered timepoint index

Primary mixed-effects model

```
Linear mixed-effects model fit by REML
  Data: data
  Log-restricted-likelihood: -281.4249
  Fixed: fixed_formula
      (Intercept)  time_factorWeek 4 time_factorWeek 12
        29.981632          -2.192893          -3.806289

Random effects:
 Formula: ~1 | patient_factor
        (Intercept) Residual
StdDev:    5.399072  1.08796

Correlation Structure: AR(1)
```

```
 Formula: ~time_index | patient_factor
 Parameter estimate(s):
       Phi
-0.1028485
Number of Observations: 126
Number of Groups: 42
```

Model summary

```
Linear mixed-effects model fit by REML
  Data: data
       AIC     BIC    logLik
  574.8498 591.723 -281.4249

Random effects:
 Formula: ~1 | patient_factor
        (Intercept) Residual
StdDev:    5.399072  1.08796

Correlation Structure: AR(1)
 Formula: ~time_index | patient_factor
 Parameter estimate(s):
       Phi
-0.1028485
Fixed effects:  list(fixed_formula)
                      Value Std.Error DF   t-value p-value
(Intercept)        29.981632 0.8498408 82  35.27912       0
time_factorWeek 4  -2.192893 0.2493223 82  -8.79541       0
time_factorWeek 12 -3.806289 0.2361533 82 -16.11787       0
 Correlation:
                   (Intr) tm_fW4
time_factorWeek 4  -0.147
time_factorWeek 12 -0.139  0.474

Standardized Within-Group Residuals:
         Min           Q1          Med           Q3          Max
-2.232442874 -0.484035640 -0.007433489  0.514088338  1.735702163

Number of Observations: 126
Number of Groups: 42
```

Coefficients

Table 78: Coefficients (block 1 of 2)

row	Value	Std Error	DF	t-value
(Intercept)	29.982	0.850	82	35.279
time_factorWeek 4	-2.193	0.249	82	-8.795
time_factorWeek 12	-3.806	0.236	82	-16.118

Table 79: Coefficients (block 2 of 2)

row	p-value
(Intercept)	0
time_factorWeek 4	0
time_factorWeek 12	0

Symptom correlation parameter: -0.102848481806327

Symptom AIC: 574.849843931725

Symptom BIC: 591.722950063959

Symptom logLik: -281.424921965862

Converged: TRUE

Comparison

Table 80: Comparison (block 1 of 2)

model	correlation structure	symptom AIC	symptom BIC
compound_symmetry	compound symmetry	575.063	591.937
ar1	AR(1)	574.850	591.723

Table 81: Comparison (block 2 of 2)

model	symptom logLik	converged
compound_symmetry	-281.532	TRUE
ar1	-281.425	TRUE

3.11.3 GEE models

Independent working correlation

Primary mixed-effects model

```
Call:
geepack::geeglm(formula = fixed_formula, family = stats::gaussian(),
    data = data, id = patient_factor, waves = time_index, corstr = correlation,
    std.err = "san.se")

 Coefficients:
                  Estimate Std.err   Wald Pr(>|W|)
(Intercept)         29.9816  0.8210 1333.6   <2e-16 ***
time_factorWeek 4   -2.1929  0.1649  176.8   <2e-16 ***
time_factorWeek 12  -3.8063  0.2347  263.0   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = independence
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)    29.66   6.015
Number of clusters:   42  Maximum cluster size: 3
```

Model summary

```
Call:
geepack::geeglm(formula = fixed_formula, family = stats::gaussian(),
    data = data, id = patient_factor, waves = time_index, corstr = correlation,
    std.err = "san.se")

 Coefficients:
```

```
                   Estimate Std.err Wald Pr(>|W|)
(Intercept)          29.982   0.821 1334   <2e-16 ***
time_factorWeek 4    -2.193   0.165  177   <2e-16 ***
time_factorWeek 12   -3.806   0.235  263   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = independence
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)     29.7    6.01
Number of clusters:   42  Maximum cluster size: 3
```

Coefficients

Table 82: Coefficients (block 1 of 2)

term	symptom Estimate	symptom robust se	symptom conf low	symptom conf high
(Intercept)	29.98	0.821	28.37	31.59
time_factorWeek 4	-2.19	0.165	-2.52	-1.87
time_factorWeek 12	-3.81	0.235	-4.27	-3.35

Table 83: Coefficients (block 2 of 2)

term	symptom Wald	symptom Pr(>|W|)
(Intercept)	1334	<0.001
time_factorWeek 4	177	<0.001
time_factorWeek 12	263	<0.001

Wald

Table 84: Wald

Df	X2	P(>|Chi|)
2	512	0

Effect tests

Table 85: Effect tests

question	symptom statistic	symptom df	symptom p raw
TIME	512	2	<0.001

Qic: 3743.47085069773, 3743.47085069773, -1868.73542534886, 2.99999999999999, 3, 3744.1024296451

Correlation structure: independence

Waves: time_index (ordered timepoint index)

Model formula

```
value ~ time_factor
<environment: 0x00000141ad785a50>
```

Converged: TRUE

Exchangeable working correlation

Primary mixed-effects model

```
Call:
geepack::geeglm(formula = fixed_formula, family = stats::gaussian(),
    data = data, id = patient_factor, waves = time_index, corstr = correlation,
    std.err = "san.se")

 Coefficients:
                   Estimate Std.err Wald Pr(>|W|)
(Intercept)          29.982   0.821 1334   <2e-16 ***
time_factorWeek 4    -2.193   0.165  177   <2e-16 ***
time_factorWeek 12   -3.806   0.235  263   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = exchangeable
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)     29.7    6.01
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha    0.959 0.00958
Number of clusters:   42  Maximum cluster size: 3
```

Model summary

```
Call:
geepack::geeglm(formula = fixed_formula, family = stats::gaussian(),
    data = data, id = patient_factor, waves = time_index, corstr = correlation,
    std.err = "san.se")

 Coefficients:
                   Estimate Std.err Wald Pr(>|W|)
(Intercept)          29.982   0.821 1334   <2e-16 ***
time_factorWeek 4    -2.193   0.165  177   <2e-16 ***
time_factorWeek 12   -3.806   0.235  263   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = exchangeable
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)     29.7    6.01
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha    0.959 0.00958
Number of clusters:   42  Maximum cluster size: 3
```

Coefficients

Table 86: Coefficients (block 1 of 2)

term	symptom Estimate	symptom robust se	symptom conf low	symptom conf high
(Intercept)	29.98	0.821	28.37	31.59
time_factorWeek 4	-2.19	0.165	-2.52	-1.87
time_factorWeek 12	-3.81	0.235	-4.27	-3.35

Table 87: Coefficients (block 2 of 2)

term	symptom Wald	symptom Pr(>|W|)
(Intercept)	1334	<0.001
time_factorWeek 4	177	<0.001
time_factorWeek 12	263	<0.001

Wald

Table 88: Wald

Df	X2	P(>|Chi|)
2	512	0

Effect tests

Table 89: Effect tests

question	symptom statistic	symptom df	symptom p raw
TIME	512	2	<0.001

Qic: 3743.47085069773, 3743.47085069773, -1868.73542534886, 2.99999999999999, 3, 3744.55193177881

Correlation structure: exchangeable

Waves: time_index (ordered timepoint index)

Model formula

```
value ~ time_factor
<environment: 0x00000141ad785a50>
```

Converged: TRUE

AR(1) correlation

Primary mixed-effects model

```
Call:
geepack::geeglm(formula = fixed_formula, family = stats::gaussian(),
    data = data, id = patient_factor, waves = time_index, corstr = correlation,
    std.err = "san.se")

 Coefficients:
                  Estimate Std.err Wald Pr(>|W|)
(Intercept)         29.982   0.821 1334   <2e-16 ***
time_factorWeek 4   -2.193   0.165  177   <2e-16 ***
time_factorWeek 12  -3.806   0.235  263   <2e-16 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = ar1
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)     29.7    6.01
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha    0.977 0.00803
Number of clusters:   42  Maximum cluster size: 3
```

Model summary

```
Call:
geepack::geeglm(formula = fixed_formula, family = stats::gaussian(),
    data = data, id = patient_factor, waves = time_index, corstr = correlation,
    std.err = "san.se")

 Coefficients:
                   Estimate Std.err Wald Pr(>|W|)
(Intercept)          29.982   0.821 1334   <2e-16 ***
time_factorWeek 4    -2.193   0.165  177   <2e-16 ***
time_factorWeek 12   -3.806   0.235  263   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = ar1
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)     29.7    6.01
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha    0.977 0.00803
Number of clusters:   42  Maximum cluster size: 3
```

Coefficients

Table 90: Coefficients (block 1 of 2)

term	symptom Estimate	symptom robust se	symptom conf low	symptom conf high
(Intercept)	29.98	0.821	28.37	31.59
time_factorWeek 4	-2.19	0.165	-2.52	-1.87
time_factorWeek 12	-3.81	0.235	-4.27	-3.35

Table 91: Coefficients (block 2 of 2)

term	symptom Wald	symptom Pr(>|W|)
(Intercept)	1334	<0.001
time_factorWeek 4	177	<0.001
time_factorWeek 12	263	<0.001

Wald

Table 92: Wald

Df	X2	P(>|Chi|)
2	512	0

Effect tests

Table 93: Effect tests

question	symptom statistic	symptom df	symptom p raw
TIME	512	2	<0.001

Qic: 3743.47085069773, 3743.47085069773, -1868.73542534886, 3, 3, 3744.55193177881

Correlation structure: ar1

Waves: time_index (ordered timepoint index)

Model formula

```
value ~ time_factor
<environment: 0x00000141ad785a50>
```

Converged: TRUE

3.11.4 Descriptive model comparison

Table 94: Descriptive model comparison (block 1 of 5)

model	formula
mixed_random_intercept	value ~ time_factor + (1 | patient_factor)
mixed_random_slope	value ~ time_factor + (1 + time_ordinal | patient_factor)
nlme_compound_symmetry	value ~ time_factor
nlme_ar1	value ~ time_factor
gee_independence	value ~ time_factor
gee_exchangeable	value ~ time_factor
gee_ar1	value ~ time_factor

Table 95: Descriptive model comparison (block 2 of 5)

model	correlation structure	n	n subjects	symptom AIC
mixed_random_intercept	random intercept	126	42	573
mixed_random_slope	random intercept + ordinal-time slope	126	42	575
nlme_compound_symmetry	compound symmetry	126	42	575
nlme_ar1	AR(1)	126	42	575
gee_independence	independence	126	42	–
gee_exchangeable	exchangeable	126	42	–
gee_ar1	ar1	126	42	–

Table 96: Descriptive model comparison (block 3 of 5)

model	symptom BIC	symptom logLik	symptom QIC
mixed_random_intercept	587	-282	–
mixed_random_slope	595	-281	–
nlme_compound_symmetry	592	-282	–
nlme_ar1	592	-281	–
gee_independence	–	–	3743
gee_exchangeable	–	–	3743
gee_ar1	–	–	3743

Table 97: Descriptive model comparison (block 4 of 5)

model	symptom CIC	converged	singular
mixed_random_intercept	–	TRUE	FALSE
mixed_random_slope	–	FALSE	TRUE
nlme_compound_symmetry	–	TRUE	–
nlme_ar1	–	TRUE	–
gee_independence	3	TRUE	–
gee_exchangeable	3	TRUE	–
gee_ar1	3	TRUE	–

Table 98: Descriptive model comparison (block 5 of 5)

model	object path
mixed_random_intercept	$model$fitted_model
mixed_random_slope	$advanced_models$random_slope$model
nlme_compound_symmetry	$advanced_models$nlme$compound_symmetry$model
nlme_ar1	$advanced_models$nlme$ar1$model
gee_independence	$advanced_models$gee$independence$model
gee_exchangeable	$advanced_models$gee$exchangeable$model
gee_ar1	$advanced_models$gee$ar1$model

Note: Likelihood criteria are descriptive and do not identify a uniquely true model. GEE rows use QIC/CIC and intentionally leave AIC, BIC, and logLik unavailable.

3.12 Effect sizes

3.12.1 Paired cohens dz

Table 99: Paired cohens dz

from	from label	to	to label	n	symptom cohens dz
pre	Baseline	week4	Week 4	42	-2.027
pre	Baseline	final	Week 12	42	-2.473
week4	Week 4	final	Week 12	42	-0.805

3.12.2 Paired rank biserial

Table 100: Paired rank biserial

from	from label	to	to label	n	symptom rank biserial
pre	Baseline	week4	Week 4	42	-1.000
pre	Baseline	final	Week 12	42	-1.000

from	from label	to	to label	n	symptom rank biserial
week4	Week 4	final	Week 12	42	-0.736

3.12.3 Repeated-measures ANOVA

Table 101: Repeated-measures ANOVA

effect	symptom partial eta squared	symptom generalized eta squared
time_factor	0.748	0.076

3.12.4 Friedman test

Symptom kendalls w: 0.831632653061225

3.12.5 Mixed models

Random intercept

Object

R2 conditional: 0.961643221415055

R2 marginal: 0.0746911898683172

Tidy

Table 102: Tidy

symptom marginal r2	symptom conditional r2
0.075	0.962

Random-slope mixed-effects model

Object

R2 marginal: 0.665217515887082

Tidy

Table 103: Tidy

symptom marginal r2	symptom conditional r2
0.665	–

Warnings: Can't compute random effect variances. Some variance components equal zero. Your model may suffer from singularity (see '?lme4::isSingular' and '?performance::check_singularity'). Decrease the 'tolerance' level to force the calculation of random effect variances, or impose priors on your random effects parameters (using packages like 'brms' or 'glmmTMB')., Random slopes not present as fixed effects. This artificially inflates the conditional random effect variances. Respecify the fixed effects structure of your model (add random slopes as fixed effects).

Symptom ICC: 0.958547051357388

3.13 Multiplicity families

Primary method: holm

Note: P-values are adjusted within scientifically distinct contrast families.

3.13.1 Pairwise arm comparisons by time point

Primary method: holm

Available adjustments: raw, bonferroni, holm, BH, BY

Object path: $advanced_tests$emmeans$time$pairwise_tidy

3.13.2 Baseline vs followup

Primary method: holm

Available adjustments: raw, bonferroni, holm, BH, BY

Object path: $advanced_tests$emmeans$baseline_followup$tidy

3.13.3 Consecutive time

Primary method: holm

Available adjustments: raw, bonferroni, holm, BH, BY

Object path: $advanced_tests$emmeans$consecutive$tidy

3.13.4 Ordinal trends

Primary method: holm

Available adjustments: raw, bonferroni, holm, BH, BY

Object path: $advanced_tests$emmeans$trends$tidy

3.13.5 Friedman posthoc

Primary method: holm

Available adjustments: raw, bonferroni, holm, BH, BY

Object path: $advanced_tests$friedman$posthoc$tidy

3.14 Robust inference

3.14.1 Cluster-robust CR2 inference

Covariance

Table 104: Covariance

row	(Intercept)	time factorWeek 4	time factorWeek 12
(Intercept)	0.690	-0.017	0.024
time_factorWeek 4	-0.017	0.028	-0.006
time_factorWeek 12	0.024	-0.006	0.056

Covariance estimator: CR2

Coefficient tests

Table 105: Coefficient tests (block 1 of 2)

Coef	beta	SE	null value	tstat
(Intercept)	29.98	0.831	0	36.1
time_factorWeek 4	-2.19	0.167	0	-13.1
time_factorWeek 12	-3.81	0.238	0	-16.0

Table 106: Coefficient tests (block 2 of 2)

Coef	df Satt	p Satt
(Intercept)	41	<0.001
time_factorWeek 4	41	<0.001
time_factorWeek 12	41	<0.001

Confidence intervals

Table 107: Confidence intervals (block 1 of 2)

Coef	beta	SE	df	CI L
(Intercept)	29.98	0.831	41	28.30
time_factorWeek 4	-2.19	0.167	41	-2.53
time_factorWeek 12	-3.81	0.238	41	-4.29

Table 108: Confidence intervals (block 2 of 2)

Coef	CI U
(Intercept)	31.66
time_factorWeek 4	-1.86
time_factorWeek 12	-3.33

Effect tests

TIME

Table 109: TIME

test	Fstat	delta	df num	df denom	p val
HTZ	244	0.976	2	40	<0.001

Effect diagnostics

TIME

Requested coefficients: time_factorWeek 4, time_factorWeek 12

Tested coefficients: time_factorWeek 4, time_factorWeek 12

Hypothesis reduced: FALSE

Test: Satterthwaite coefficient tests; HTZ joint Wald tests

3.15 Sensitivity analyses

Table 110: Sensitivity analyses (block 1 of 3)

question	method	n	symptom effect size	symptom statistic
TIME	Mixed model: random-intercept LRT	126	–	115.8
TIME	RM-ANOVA (uncorrected)	42	0.748	121.7
TIME	RM-ANOVA (Greenhouse-Geisser)	42	0.748	121.7
TIME	RM-ANOVA (Huynh-Feldt)	42	0.748	121.7
TIME	Friedman test	42	0.832	69.9
TIME	GEE (independence)	126	–	512.3

question	method	n	symptom effect size	symptom statistic
TIME	GEE (exchangeable)	126	–	512.3
TIME	GEE (ar1)	126	–	512.3
TIME	Mixed model: random-slope LRT	126	–	84.8
TIME	Mixed model: CR2/HTZ robust Wald test	126	–	244.0

Table 111: Sensitivity analyses (block 2 of 3)

question	symptom df	symptom p raw	effect size type
TIME	2	<0.001	–
TIME	2/82	<0.001	partial eta squared
TIME	1.39909823858027/57.3630277817911	<0.001	partial eta squared
TIME	1.4333543807245/58.7675296097043	<0.001	partial eta squared
TIME	2	<0.001	Kendall's W
TIME	2	<0.001	–
TIME	2	<0.001	–
TIME	2	<0.001	–
TIME	2	<0.001	–
TIME	2/40	<0.001	–

Table 112: Sensitivity analyses (block 3 of 3)

question	object path	note
TIME	$model$global_time_test	–
TIME	$advanced_tests$rm_anova$object	Complete-profile repeated-measures analysis.
TIME	$advanced_tests$rm_anova$tables$greenhouse_geisser	Degrees of freedom and p-value corrected for sphericity.
TIME	$advanced_tests$rm_anova$tables$huynh_feldt	Degrees of freedom and p-value corrected for sphericity.
TIME	$advanced_tests$friedman$test	Rank-based complete-profile global test.
TIME	$advanced_models$gee$independence$model	Robust sandwich Wald test under the stated working correlation.
TIME	$advanced_models$gee$exchangeable$model	Robust sandwich Wald test under the stated working correlation.
TIME	$advanced_models$gee$ar1$model	Robust sandwich Wald test under the stated working correlation.
TIME	$advanced_models$random_slope$global_time_test	Random slope uses the ordinal timepoint index.
TIME	$robustness$club_sandwich$effect_tests$TIME	CR2 covariance with small-sample HTZ correction.

Interpretation: Use this table to compare sensitivity across methods. It is not a majority vote and does not designate an effect as true based on the number of small p-values.

3.16 Diagnostic outliers

Note: IQR flags are diagnostic flags and are not automatically excluded from analyses.

3.16.1 By time point

Pre

Outliers detected: 0.

Week4

Outliers detected: 0.

Final

Outliers detected: 0.

3.16.2 Within-participant changes

Pre to week4

Table 113: Pre to week4

patient	symptom change	symptom lower bound	symptom upper bound
E009	-5.52	-4.38	0.094

Pre to final

Outliers detected: 0.

Week4 to final

Outliers detected: 0.

3.17 Diagnostics and adaptations

3.17.1 Automatic adaptations

Symptom

Available by time: 42, 42, 42

Variable timepoints: 3

3.18 Figures

3.18.1 Distribution by time point

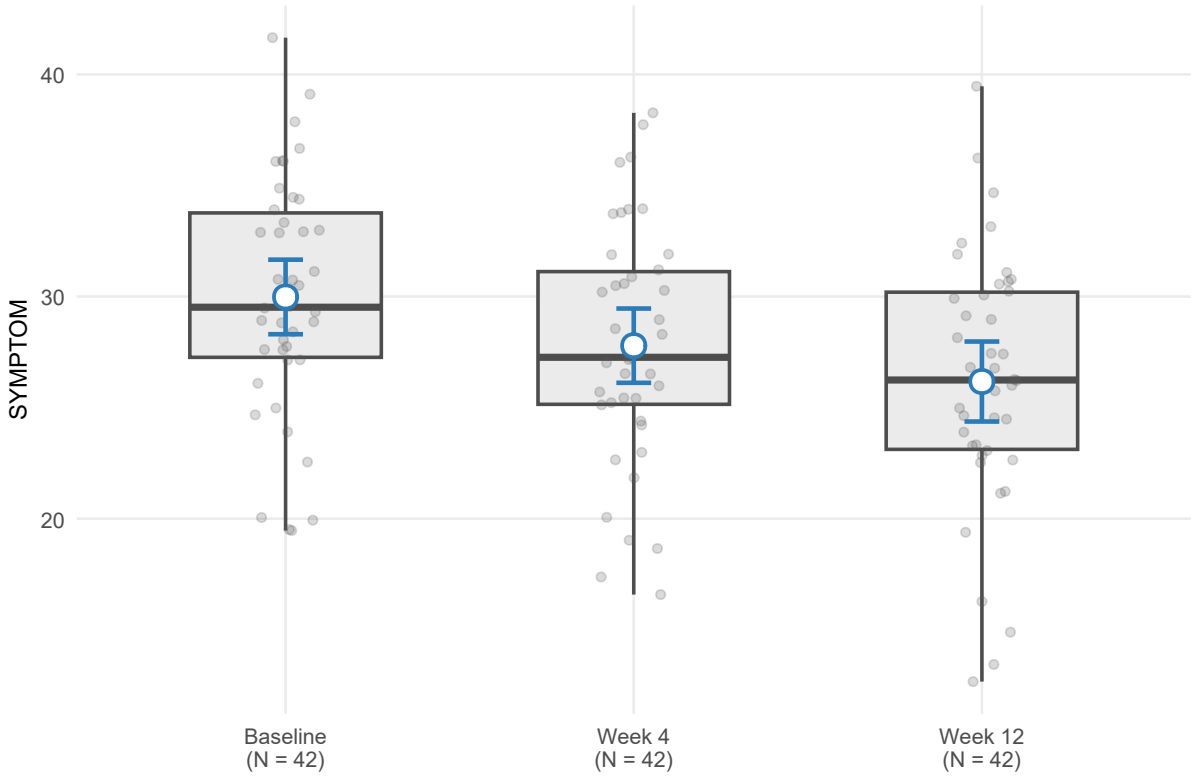


Figure. Distribution of observed values by time point, with individual observations and confidence intervals for the mean.

3.18.2 Spaghetti

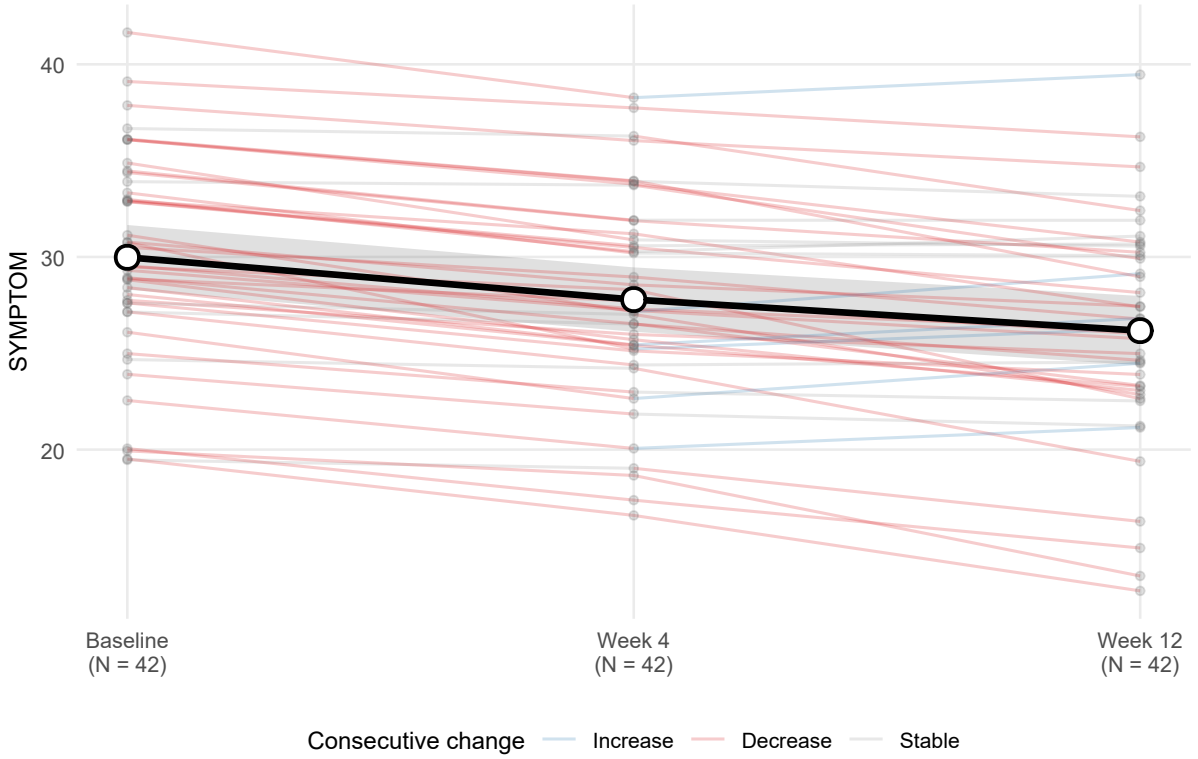


Figure. Individual longitudinal trajectories.

3.18.3 Mean profile with confidence intervals

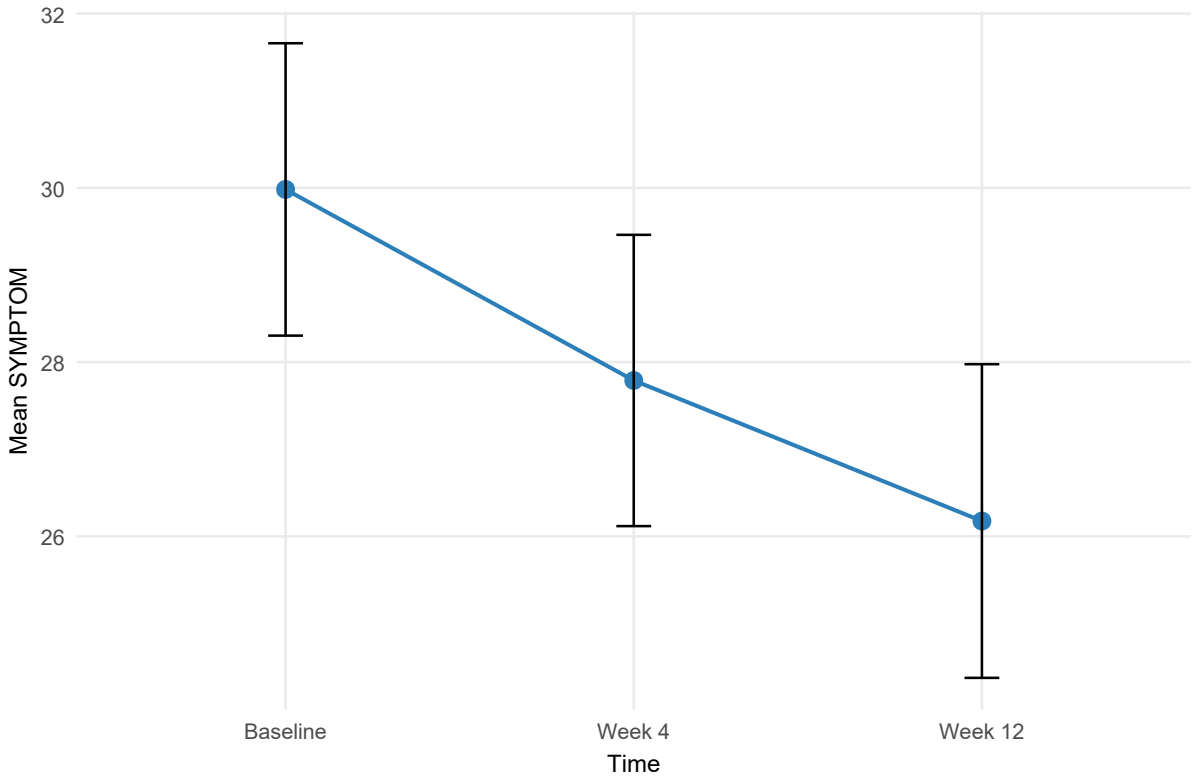


Figure. Longitudinal mean profile with confidence intervals.

3.18.4 Within-participant changes

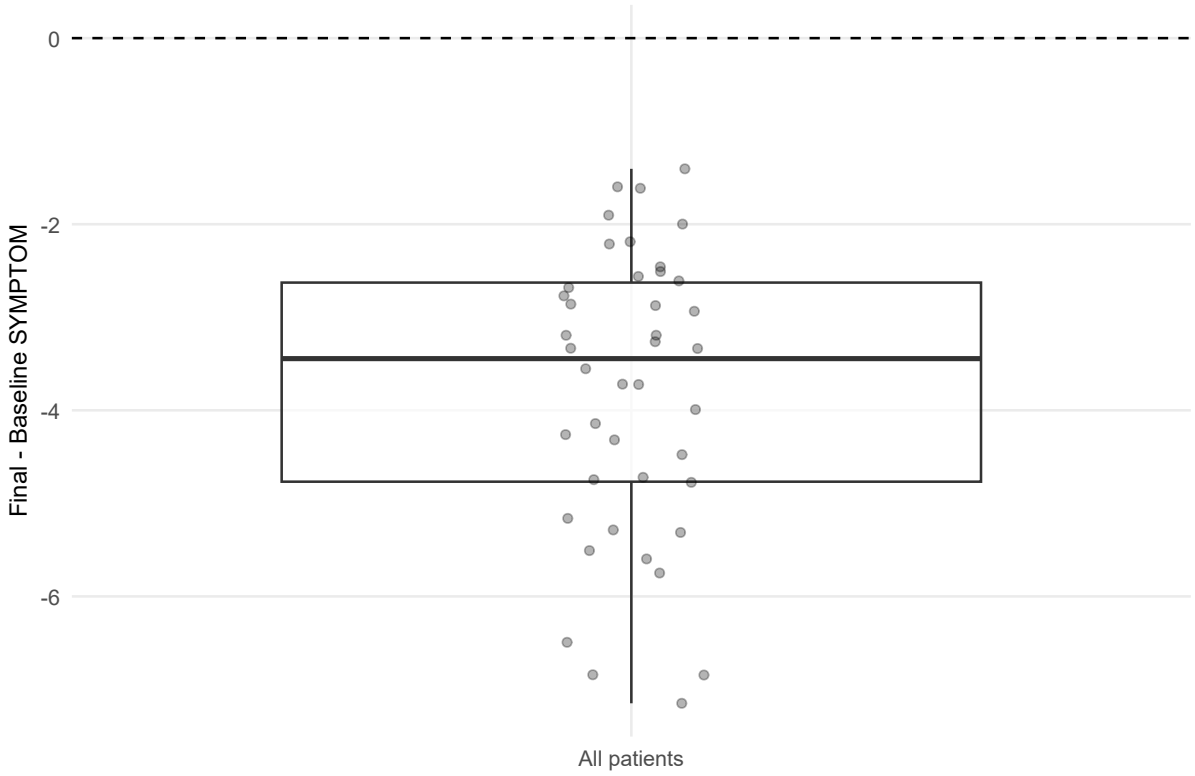


Figure. Distribution of within-participant changes.

3.18.5 Change from baseline

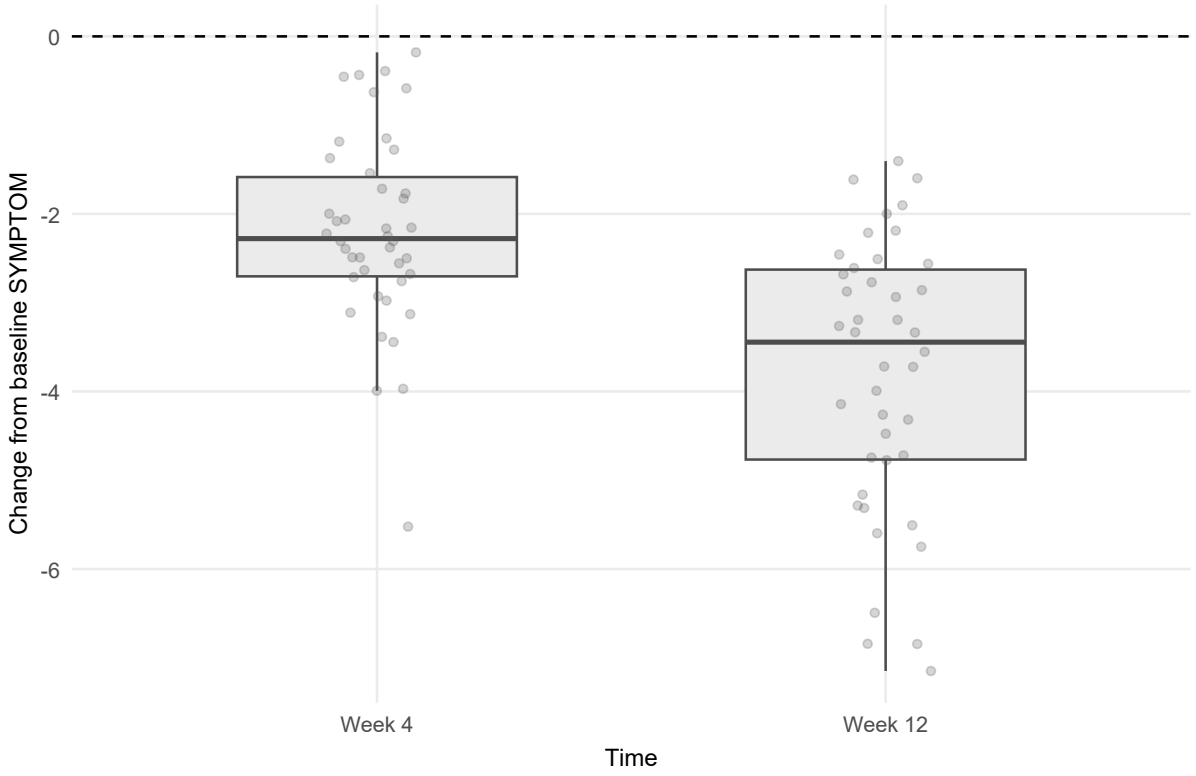


Figure. Change from baseline during follow-up.

3.18.6 Mean change with confidence intervals

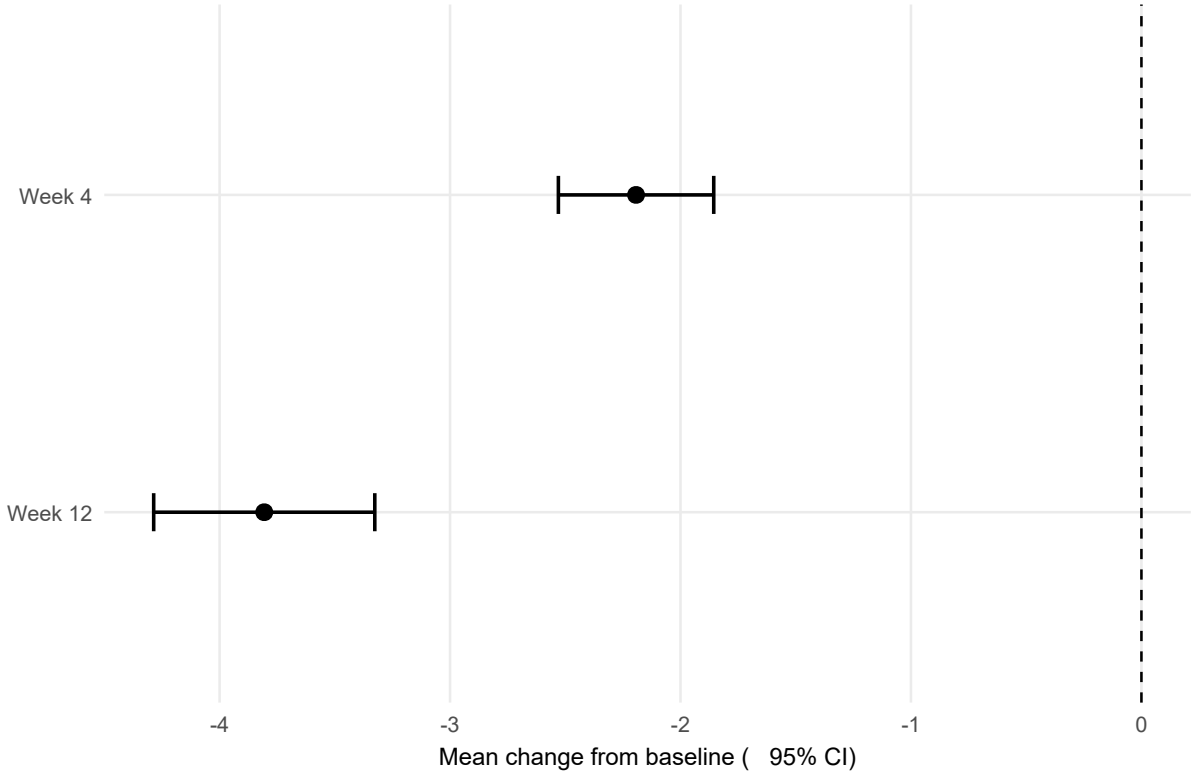


Figure. Mean longitudinal change with confidence intervals.

3.18.7 Missing data

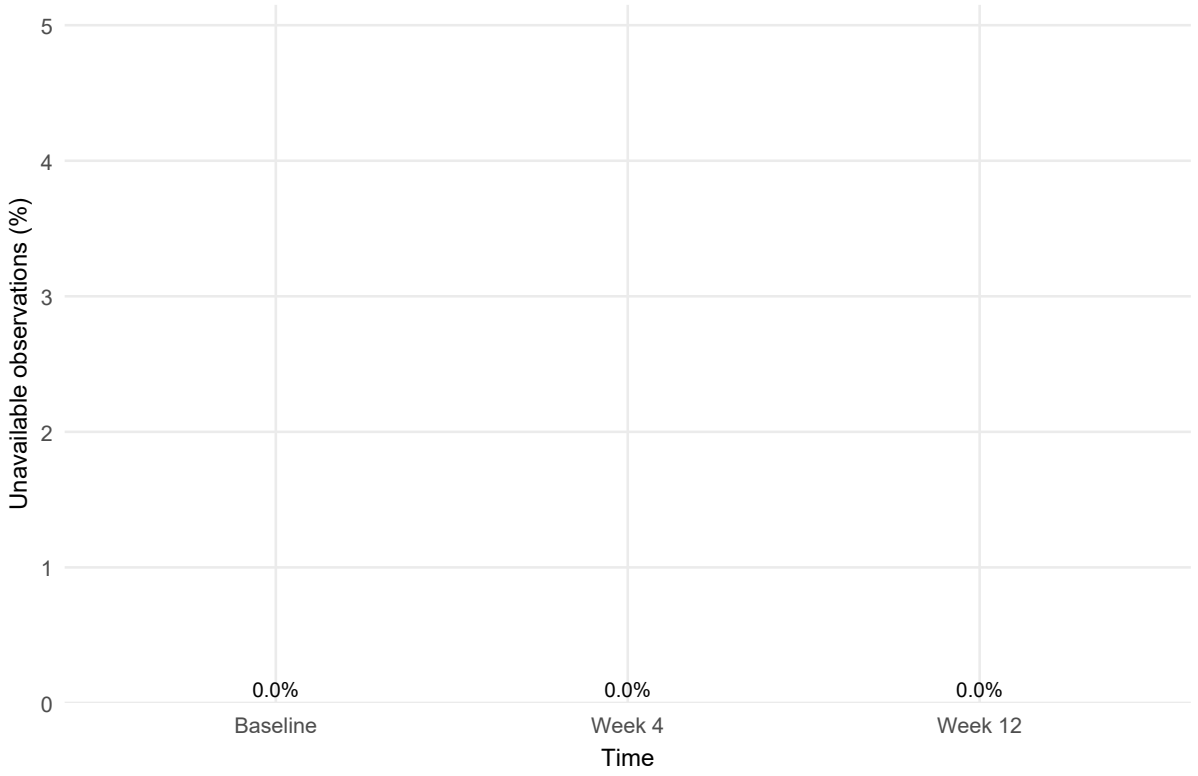


Figure. Percentage of missing observations by time point.

3.18.8 Correlation matrix

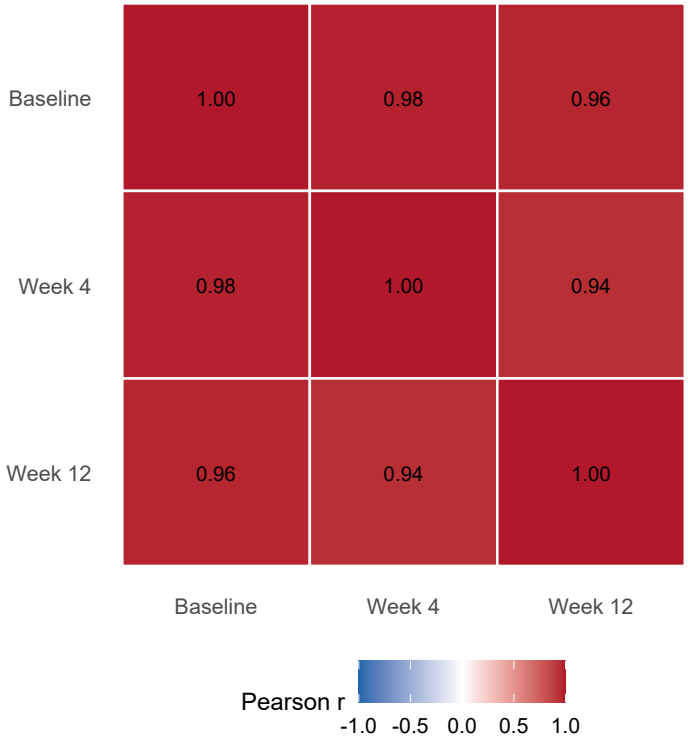


Figure. Correlation structure across time points.

3.18.9 Individual clinical response

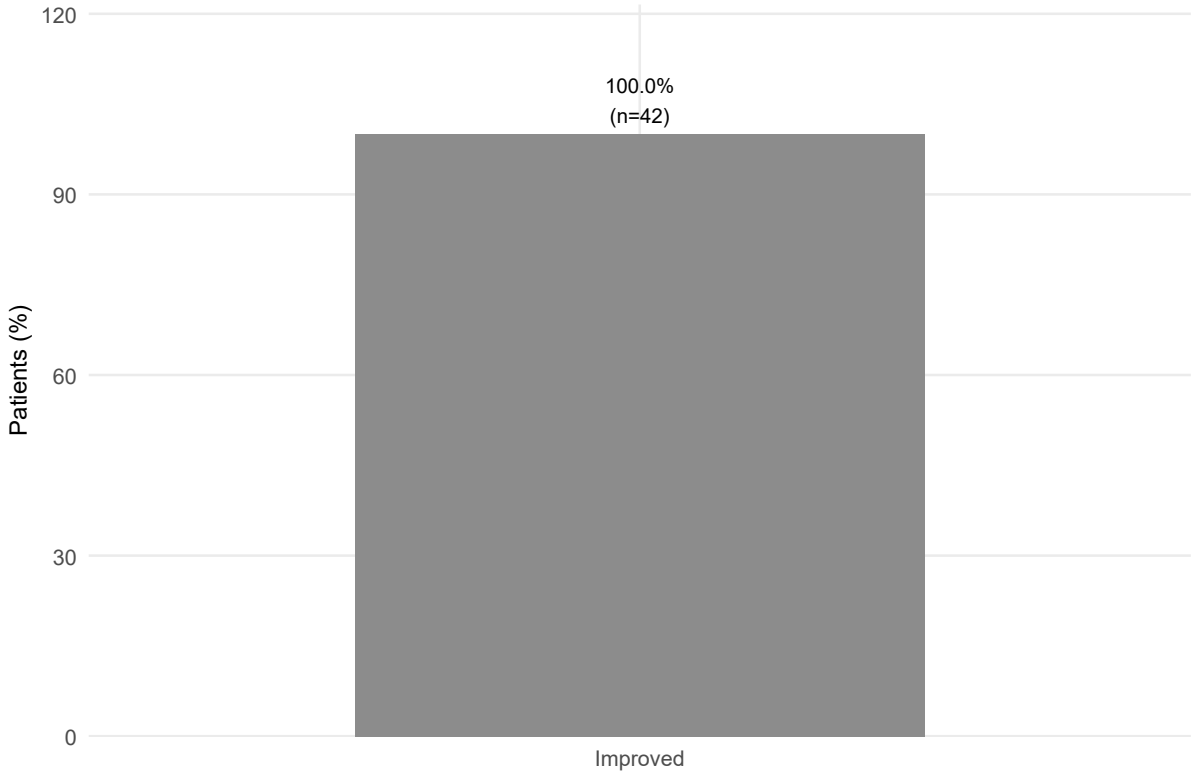


Figure. Frequency of individual response directions.

4 Descriptive summary

Table 114: Differences between observed means

outcome	mean first visit	mean last visit	descriptive difference
symptom	30	26.2	-3.81